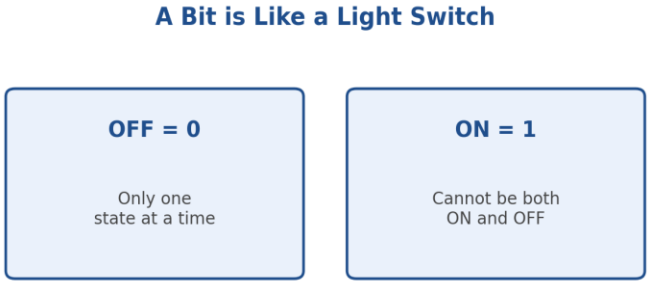
Bits, Qubits and Quantum States

Why Do We Need to Understand Information

Every computer works with information. Before learning quantum computing, it helps to understand how information is stored and represented. Classical computers use bits, while quantum computers use qubits. The way these two systems represent information is fundamentally different, and this difference is the first step toward understanding quantum computing.

1. Bits

A bit is the smallest unit of information in a classical computer. A bit can only hold one value at a time, either 0 or 1. Everything inside a computer, phone, or laptop is ultimately stored using combinations of these two values.



How Computers Use Bits

Computers combine multiple bits to represent larger amounts of information. Each additional bit doubles the number of possible values.

Number of Bits	Possible Values
1 Bit	2 Values
2 Bits	4 Values
3 Bits	8 Values
8 Bits	256 Values

For example: 0001 = 1, 0010 = 2, 0100 = 4, 1000 = 8. This binary representation is the foundation of classical computing, and bits are used to store text, images, videos, audio, and programs.

2. Qubits

A qubit is the basic unit of information in a quantum computer. Unlike a bit, a qubit is not restricted to only one state. It can exist in 0, 1, or a combination of both, a property that comes from quantum mechanics.

Analogy: A Spinning Coin

While a coin is spinning, both heads and tails are possible, and it is not fixed to one outcome. Similarly, a qubit can exist in a combination of states until it is measured, after which it becomes either 0 or 1.

Why Qubits Are Powerful

A single bit stores one value, while a single qubit stores information using probabilities of multiple possibilities. When multiple qubits work together, the number of possible states grows rapidly, which allows quantum computers to explore large solution spaces more efficiently for certain problems.

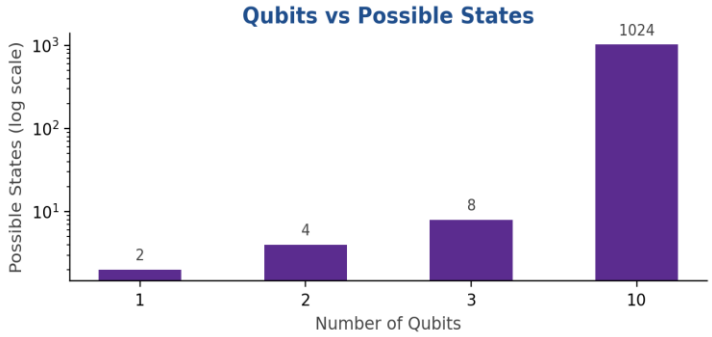


Figure 1: Each added qubit doubles the number of possible states.

Real Hardware Qubits

Qubits are not imaginary objects. Researchers create them using physical systems such as superconducting circuits, trapped ions, photons, and neutral atoms. IBM's quantum computers use superconducting qubits cooled to extremely low temperatures.

3. Quantum States

A quantum state describes the complete condition of a qubit. It tells us everything we can know about that qubit before measurement. Instead of saying only 0 or only 1, a quantum state describes the probability of obtaining either result.

State Representation

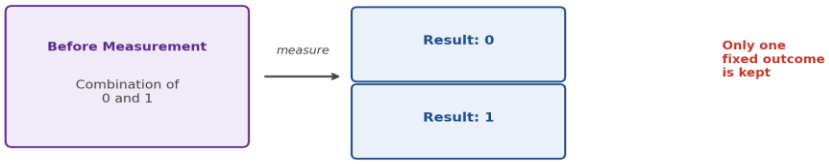
The two basic quantum states are written as |0> and |1>. These symbols are called ket notation, and they are commonly used in quantum computing and quantum mechanics.

Superposition as a Quantum State

A qubit does not always stay in only |0> or |1>; it can exist in a combination of both. For example, a state may hold a 50 percent chance of measuring 0 and a 50 percent chance of measuring 1. This is known as a superposition state, and the quantum state stores these probabilities until measurement occurs.

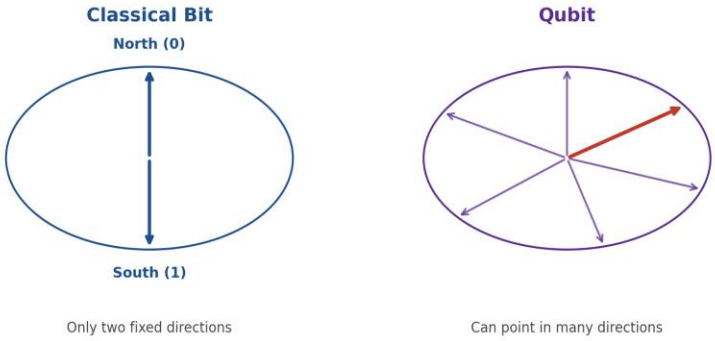
Measurement Changes the State

Before measurement, a qubit holds a combination of possibilities. After measurement, it becomes either 0 or 1, and the original quantum state collapses into a definite result. This behaviour is very different from classical bits.



Visual Understanding

Think of a compass needle. A classical bit can point only north or south, while a qubit can point in many directions between them. The direction represents its quantum state, which is why quantum states can carry richer information than classical bits.



4. Bits vs Qubits

Feature	Bit	Qubit
Values	0 or 1	0, 1, or combination
Based on	Classical physics	Quantum mechanics
State before observation	Fixed	Probabilistic
Information capacity	Limited	Richer representation
Measurement	No change	State collapse

Summary: What to Remember

- A bit is the smallest unit of classical information and can only be 0 or 1
- A qubit is the smallest unit of quantum information and can exist in multiple possibilities at once
- More qubits mean exponentially more possible states
- A quantum state is the complete description of a qubit, including its probabilities
- Ket notation, |0> and |1>, is used to represent basic quantum states
- Superposition allows a qubit to hold a combination of states until measured
- Measurement collapses a quantum state into one definite outcome
- Qubits are built using real hardware such as superconducting circuits, trapped ions, photons, and neutral atoms